

Student Tutorial 2: Creating a README for Research Code

Hands-on activity: writing a minimal but useful README

What you will learn

In this exercise, you will learn how to create a simple README file for a research project (e.g. your own project or a hypothetical project). A README is usually the first file someone reads when they open a repository. It explains what the project is, how to use it, and how to cite or contact the authors.

Why a README matters

A repository without a README can be visible but still hard to reuse. A good README helps other people answer basic questions:

- What is this project?
- What does it do?
- Which file should I open first?
- How do I run it?
- What software does it need?
- How should I cite it or contact the authors?
- Am I allowed to reuse it?

Your task

Create or improve a file called README.md for a small research project.

Minimum README contents

- project name
- short description
- files included
- how to run
- dependencies
- data
- citation or contact
- license

Example template

```
# Project name
```

```
## Short description of the project.
```

```
Explain the purpose of the project in 2-3 sentences.
```

Files included

- analysis.py: main analysis script
- data/README.txt: information about the data

How to run

Explain the basic steps needed to run the code.

Example:

```
python analysis.py data/example.csv results/summary.txt
```

Dependencies

List any software or packages needed. Specific versions and dependencies.

Data

Explain whether the data are included, restricted, synthetic, or available elsewhere.

Citation or contact

Explain how this project should be cited, or provide contact details.

License

State the license, if one has been chosen.

A vague README versus a useful README

Too vague:

This repository contains scripts.

More useful:

This repository contains a small Python script that reads an example dataset, summarizes the number of rows, and writes a text output file.

Questions to guide your writing

- Would a colleague understand what this project is for?
- Would they know which file to open first?
- Would they know what they need to install?
- Would they know how to run a minimal example?
- Would they understand whether data are available?
- Would they know how to cite or contact you?
- Would they understand whether they are allowed to reuse the code?

Reflection question

If someone found this repository online, would they understand what it is and how to start within five minutes?

Optional extension: citation metadata

For research code, a README can point to a release, DOI, paper, or CITATION.cff file. This helps users cite a specific version rather than only linking to the latest repository state.